The Development and Applications of BioImpedance Technology

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Introduction of Bioimpedance Technology

> 💡 生物阻抗技術是把**歐姆定律**應用到生物組織上，分析組織的電性特徵。
> 不同組織（例如肺、心臟、脂肪、肌肉、病變組織）的導電率與阻抗不同，因此可藉由量測阻抗變化來推估組織狀態。

- Ohm's Law
  - $\boldsymbol{J} = \sigma \boldsymbol{E}$, where $\boldsymbol{J}$ is a vector form of current density, and $\boldsymbol{E}$ is a vector form of electric field intensity.
  - V = IR, where V is the voltage difference, I the current, and R the resistance.

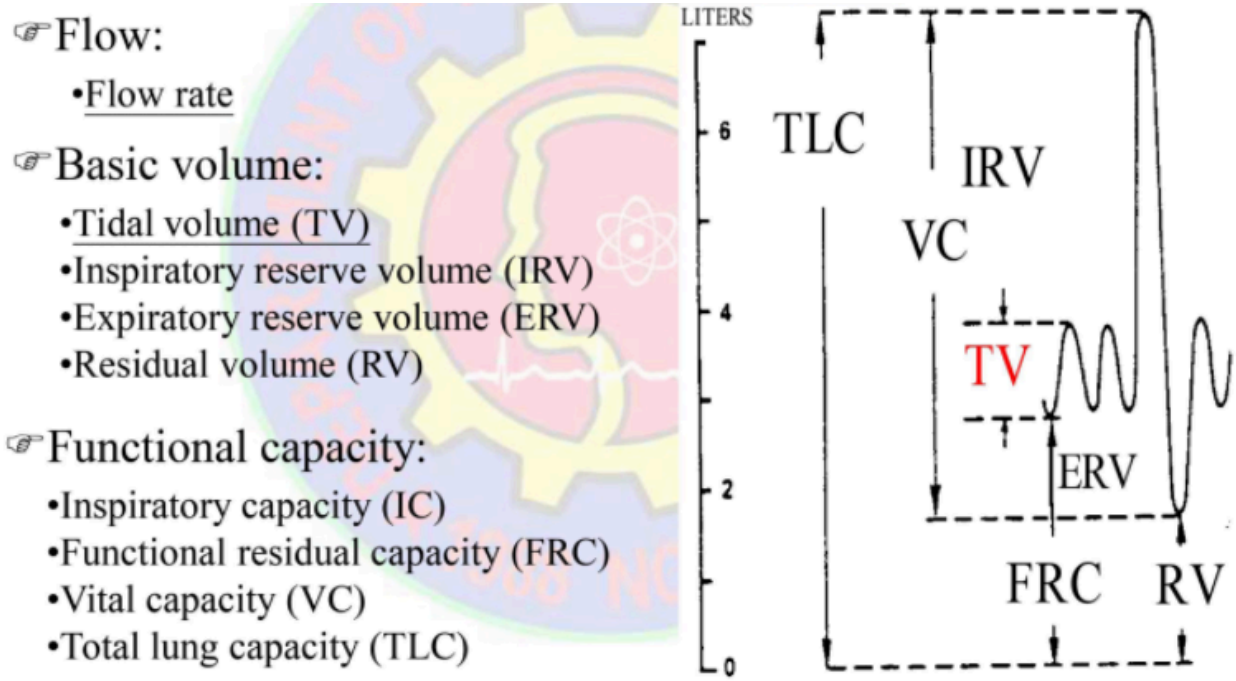
電磁學中：電流密度與電場關係

$$J = \sigma E$$

電路學中：電壓、電流、電阻關係

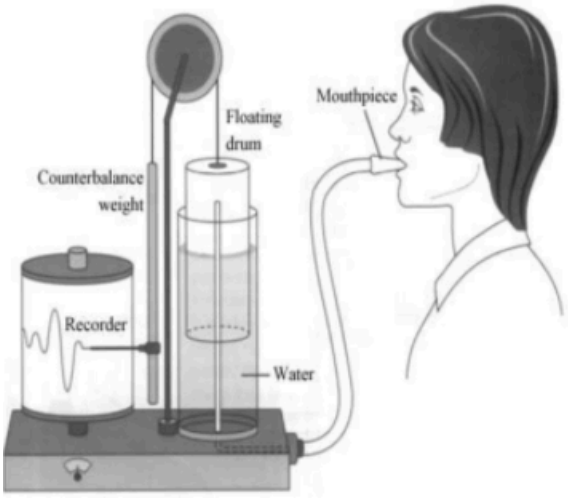
$$V = IR$$

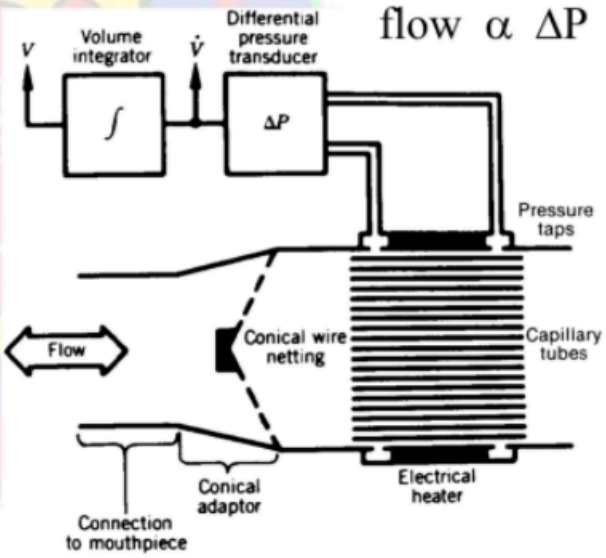
Physiological Parameters of Respiration



在呼吸與循環上的應用

☞Respiratory volume and flow meters:
spirometer, rotating-vane, ultrasonic, and differential pressure flow meter.





呼吸：

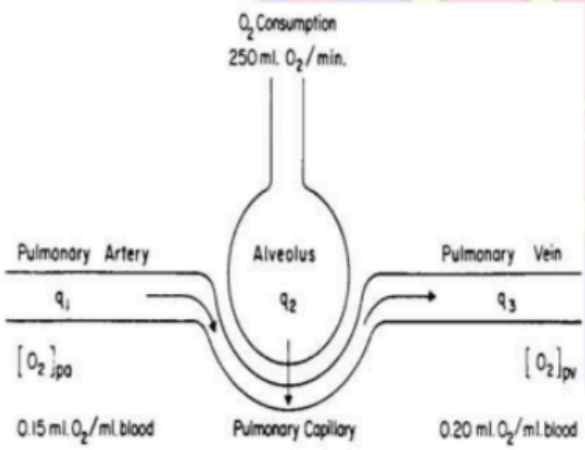
- 潮氣量（TV）
- 吸氣儲備量（IRV）
- 呼氣儲備量（ERV）
- 殘氣量（RV）
- 吸氣容量 (IC)
- 功能性殘氣量 (FRC)
- 肺活量（VC）
- 總肺容量（TLC）
- 流量變化

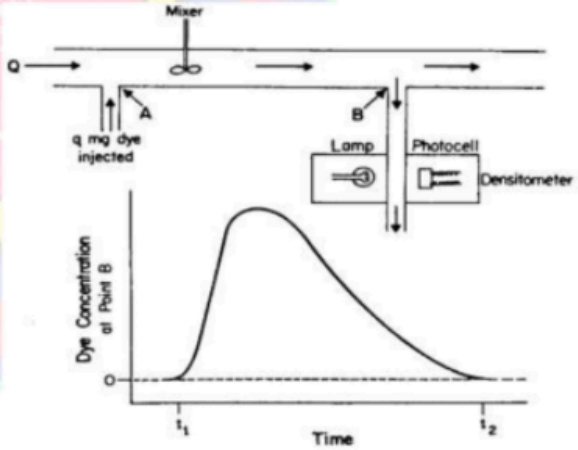
❑Cardiac output (CO):

☞ $CO = SV \times HR$, where SV denotes stroke volume, and HR heart rate.

❑Methods:

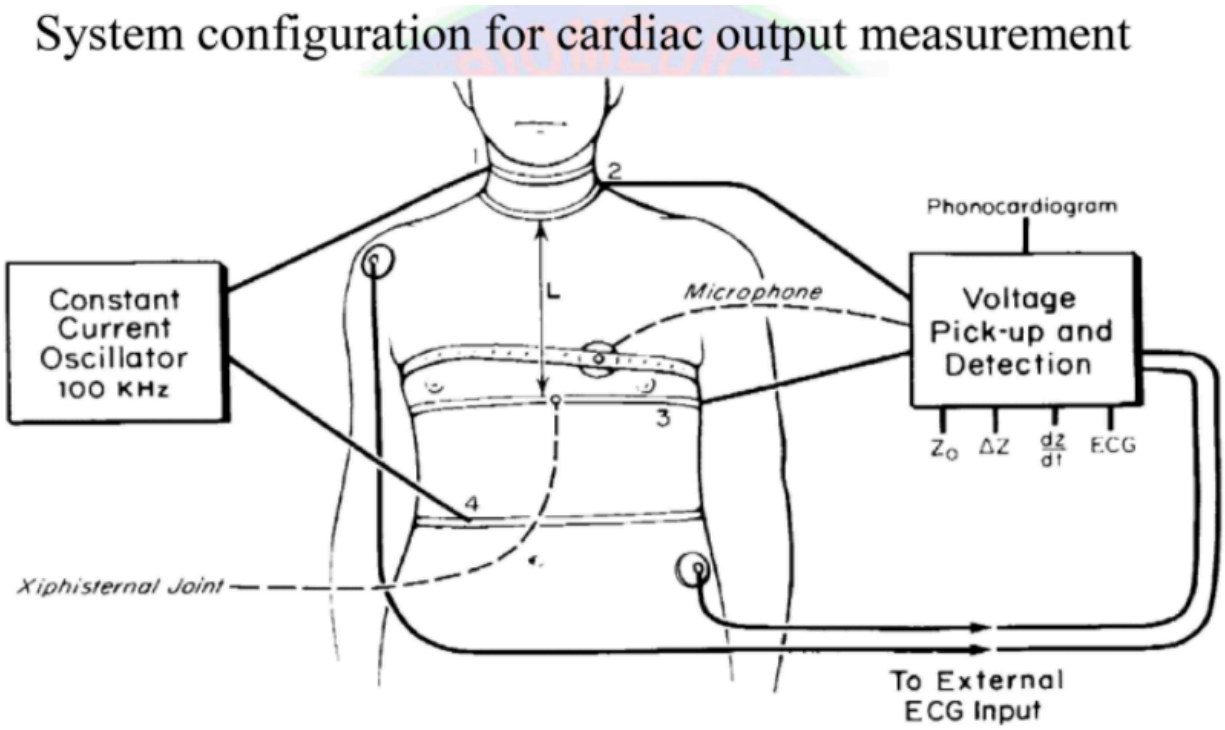
☞Oxygen Fick method, Indicator dilution method, etc.





循環：

- **心輸出量（Cardiac Output, CO）**
  - SV：每次脈搏輸出量
  - HR：心率



Bioimpedance 的應用

- **體積變化量測**
  - Impedance plethysmography
  - Impedance cardiography
  - Impedance pneumography
- **身體組成分析**
- **組織分類**
- **組織監測**
- **細胞層級測量**

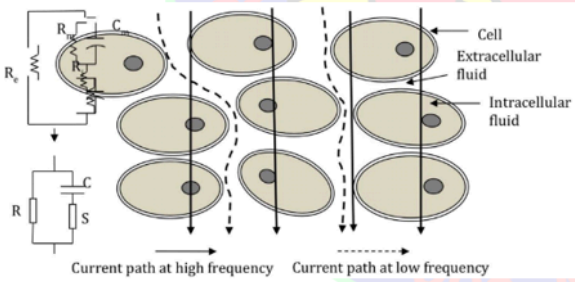
生物組織的阻抗可寫成：

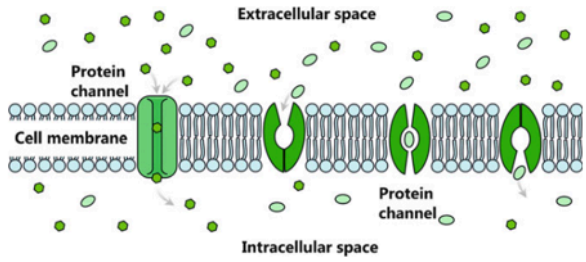
$$Z(\omega) = R(\omega) + jX(\omega) = |Z(\omega)|e^{j\phi(\omega)}$$

測量生物使用：

- 交流電

- 頻率 大於 10k Hz
- 加電流，測電壓



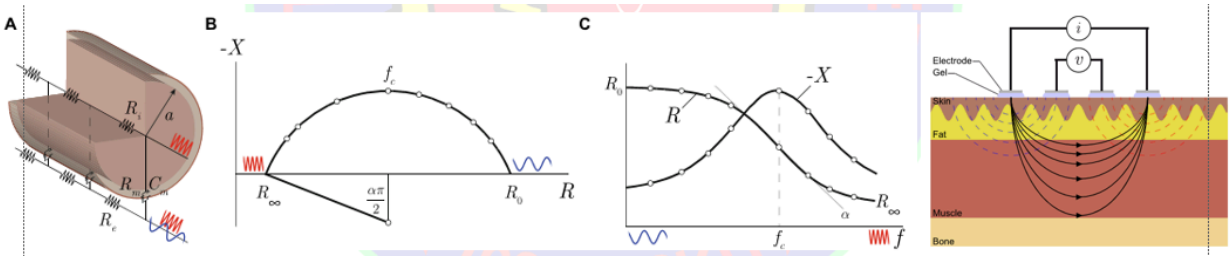


A. Stahn, E. Terblanche, and H. Gunga, Chapter 3 in V. R. Preddy (ed.) Handbook of Anthropometry: Physical Measures of Human Form in Health and Disease, Springer, 2012.

Electrical impedance myography (EIM)

主要價值：

- 評估疾病的嚴重程度
- 疾病惡化
- 治療反應等



Overview of Electrical Impedance Tomography (EIT)

特點：

- 非侵入性
- 無輻射

- 即時成像系統

不同成像方式的比較

Medical imaging modality	CT	US	MRI	EIT
Basic principle	X-rays	High frequency sound	Radio waves	Impedance
Types of radiation	Ionizing radiation	Non-Ionizing radiation	Non-Ionizing radiation	Non-Ionizing radiation
Contrast	High	Low	High	Low
Spatial Resolution	50-200 μm	50-500 μm	25-100 μm	Low
Scanning time	<20 min	< 30 min	<40 min	<10 min
Cost	Moderate	Low	Very High	Low
Size	Non portable	Portable	Non portable	Portable
Advantages	Bone and tumor imaging, anatomic imaging	Visualize muscles, tendon and internal organs	Morphological and functional imaging	Rapid tomographic imaging, low cost, noninvasive
Disadvantages	High cost Ionizing radiation	Operator dependency	Noisy, cost, low sensitivity	Not mature yet

成像系統的主要組成部分

- Energy source
- Tissue-energy interaction
- Measurable quantity
  - detectability
- Reconstruction algorithm
  - governing equation

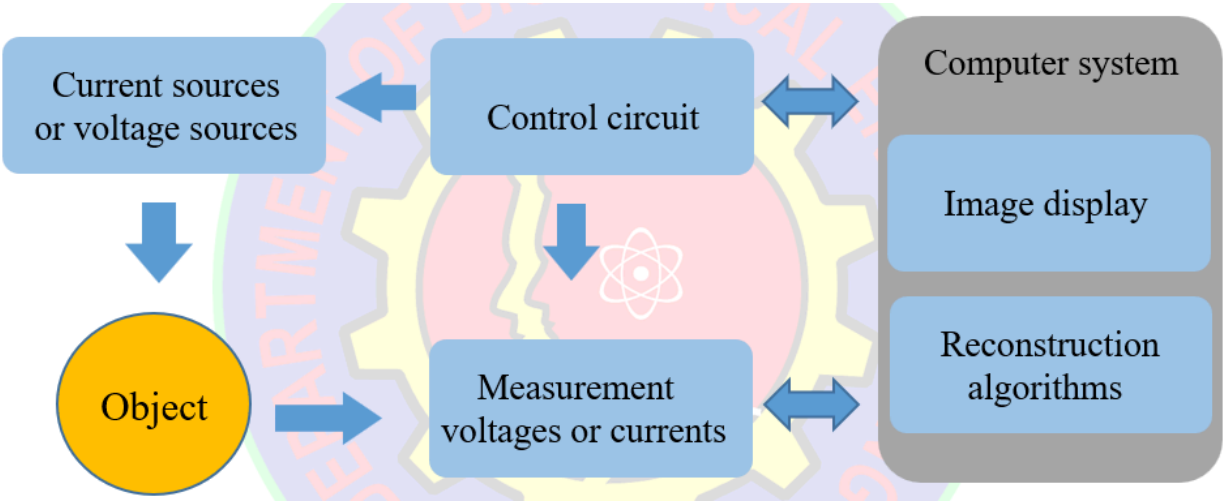
顯示訊息

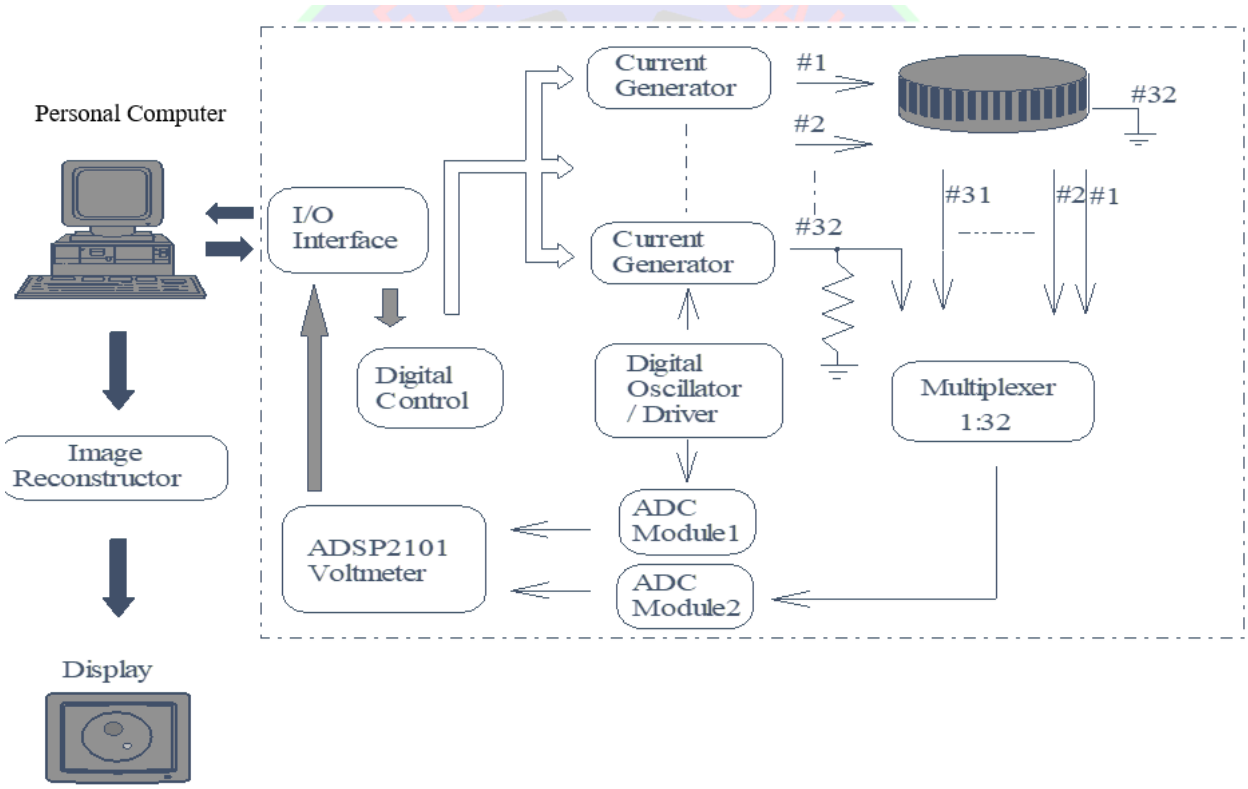
- Anatomical image 解剖影像
- Functional image 功能影像

- Molecular image 分子影像

System Configuration

- 電極陣列
- 電流源或電壓源
- 控制電路
- 量測模組
- 電腦系統
- 影像重建演算法
- 影像顯示





核心流程

1. 在體表放置多個電極
2. 注入微小交流電流
3. 量測各電極的電壓反應
4. 利用演算法推回體內阻抗分布
5. 重建影像

市面上可用於臨床的幾種 EIT 系統

- Pulmovista 500 (Dräger Medical GmbH, Lündbeck, Germany)
- Goe-MF II (CareFusion - Becton, Dickinson and Company, San Diego, United States)
- Mark 1 and Mark 3.5(Maltron International Ltd, Rayleigh, United Kingdom)
- BB (Swisstom AG, Landquart, Switzerland)
- Enlight (Timpel S/A, Sao Paulo, Brazil)

Image Reconstruction Algorithm

- **Forward problem**

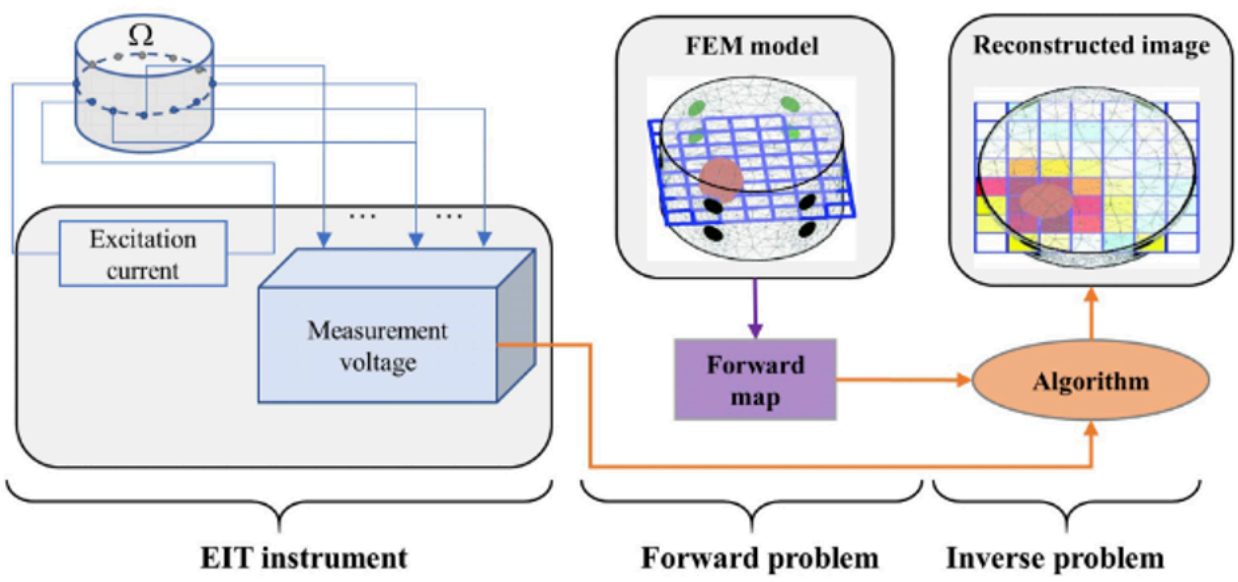
Given internal resistivity ρ
boundary current density J_o → **Forward Solver** → Find voltage V, and J

Solver: Analytic method, FEM , ...

- **Inverse problem**

Given boundary voltage V_o
and current density J_o → **Inverse Solver** → Find internal resistivity

Solver: NOSER, Backprojection, GA, SA, NN, DL, ...



Forward problem

> 已知：
>
> - 內部電阻率或導電率

> - 邊界條件（施加電流）
>
> 求：
>
> - 邊界電壓
> - 電場/電流分布

Inverse problem

> 已知：
>
> - 邊界量測的電壓與電流
>
> 求：
>
> - 內部電阻率或導電率分布

💡 EIT 的 inverse problem 具有：

- **非線性**
- **ill-posed**
  ⇒
  - 小小的量測誤差會造成很大的重建誤差
  - 解可能不唯一
  - 影像容易模糊

傳統影像重建演算法

- Filtered backprojection
- Newton One-Step Estimation Reconstruction
- Simulated annealing
- Genetic algorithms

Clinical Application Examples

優勢：

- No radiation hazard
- Non-invasive technique
- Real-time and continuous measurement
- Portability for bedside monitoring

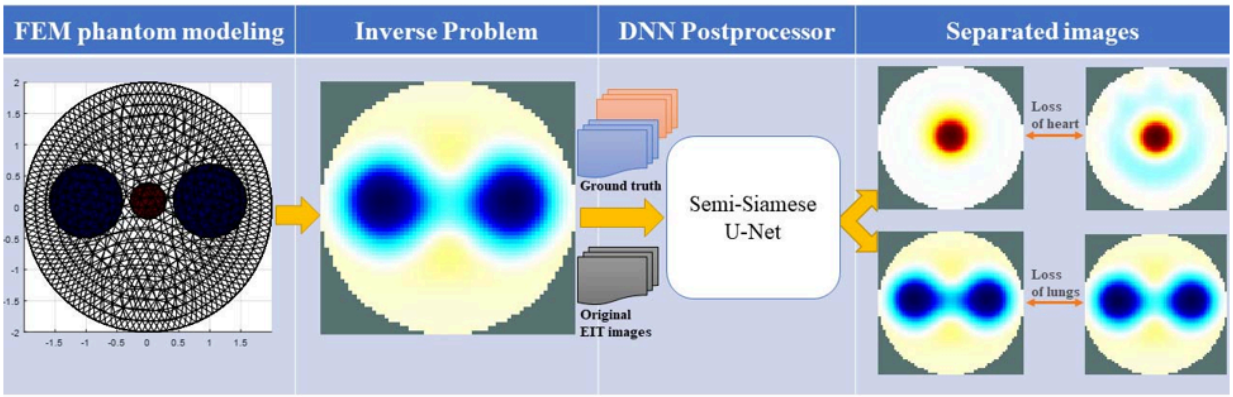
用途：

1. Dynamic evaluation of therapeutic interventions
2. Regional ventilation
3. Regional alveolar behavior
4. Optimal PEEP setting
5. Monitoring pulmonary perfusion

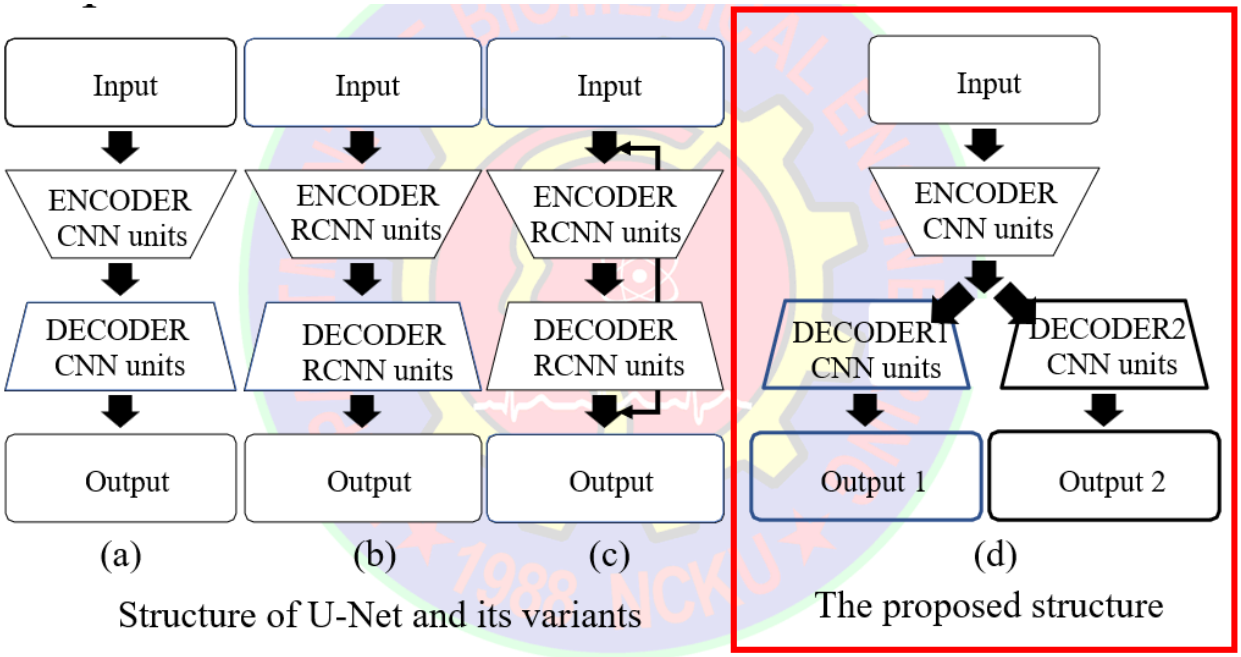
Deep Learning in Impedance Image Separation

U-Net 是醫學影像分割常見架構

1. Better U-nets
2. Attention U-nets
3. Inception U-nets
4. Ensemble U-nets



Semi-Siamese U-net Structure



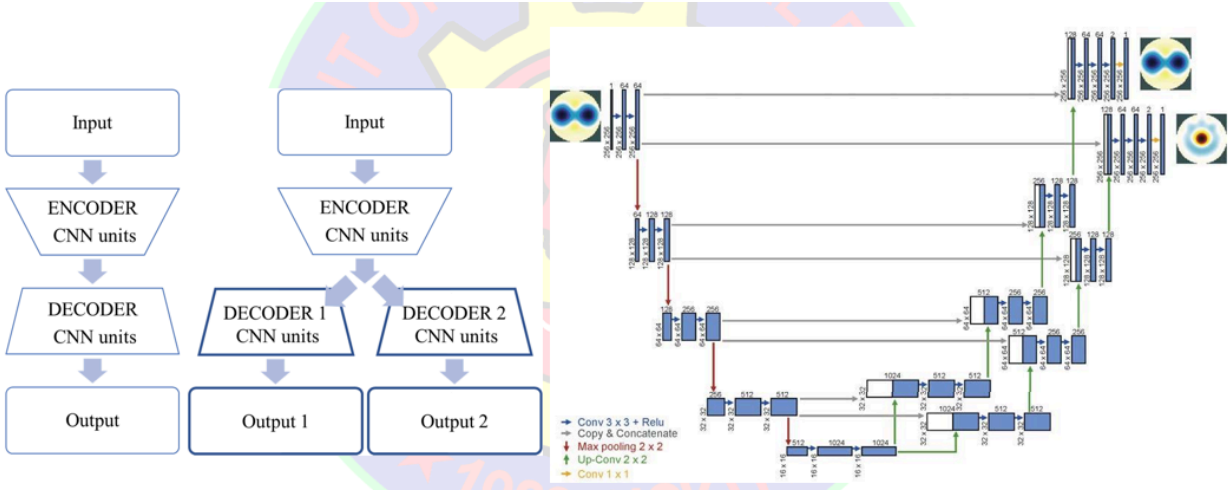
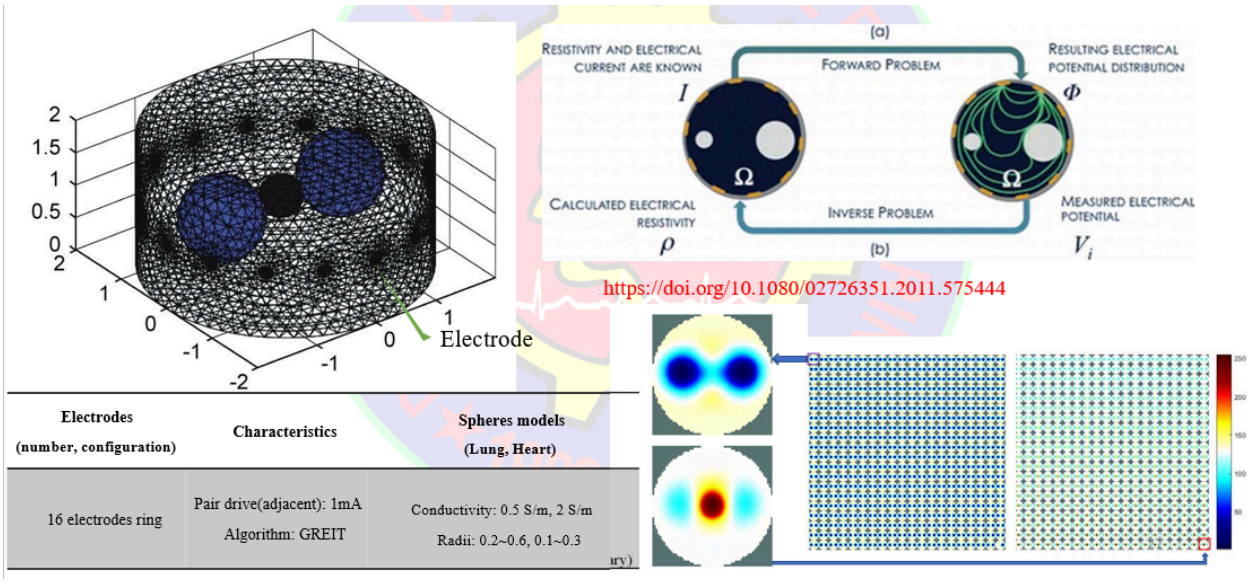


Table 1. Semi-Siamese U-Net architecture details.

Contracting Path			Expanding Paths Path1 and Path2		
Block	Layer	Filter	Block	Layer	Filter
Block1	Conv2D(3,3)	64	Block 6	Conv2D(2,2)	512
	Conv2D(3,3)	64		Conv2D(3,3)	512
	MaxPooling2D(2,2)			Conv2D(3,3)	512
	Concatenate Block9				
Block2	Conv2D(3,3)	128	Block 7	Conv2D(2,2)	256
	Conv2D(3,3)	128		Conv2D(3,3)	256
	MaxPooling2D(2,2)			Conv2D(3,3)	256
	Concatenate Block8				
Block3	Conv2D(3,3)	256	Block8	Conv2D(2,2)	128
	Conv2D(3,3)	256		Conv2D(3,3)	128
	MaxPooling2D(2,2)			Conv2D(3,3)	128
	Concatenate Bolck7				
Block4	Conv2D(3,3)	512	Block9	Conv2D(2,2)	64
	Conv2D(3,3)	512		Conv2D(3,3)	64
	Dropout			Conv2D(3,3)	64
	MaxPooling2D(2,2)			Conv2D(3,3)	2
	Concatenate Block6				
Block5	Conv2D(3,3)	1024		Conv2D(1,1)	1
	Conv2D(3,3)	1024			
	Dropout				

Table 2. The parameter of models in our experiments.

Model	Dual U-Net (original)	Semi-Siamese U-Net (proposed)
Parameters	62,063,370	43,221,322

3D 模擬肺部與心臟